

BIKE-SHARING PLATFORM

MOCK TEST

DATA FILES

You are provided with the following **CSV files**: users.csv, bikes.csv, stations.csv, rides.csv, payments.csv. All data represents a **bike-sharing system** operating in a city.

USING TOOLS

1. You can use both local tools (VS Code, Jupiter Noptebok, Microsoft Excel) and those available in the cloud (Google Colab).
2. You can use information available online.
3. **Do NOT use artificial intelligence.**

PART I – JUPYTER NOTEBOOK

Task 1. Notebook structure

Using Markdown cells, prepare the structure of the notebook:

1. Add a title of the project.
2. Add a numbered list of authors (name, surname, id).
3. Describe in 4-5 sentences what Bike-Sharing is and how it works.
4. Present in the table three advantages and three disadvantages of the BIKE SHARING PLATFORM. Add column titles in bold.
5. Complete the following tasks in Jupyter Notebook. At the beginning of each task, add its title and description (copy and format task title and points to Jupiter notebook).

Task 2. Data loading and inspection

1. Load all CSV files into separate Pandas DataFrames.
2. For each DataFrame:
 - describe (markdown) what the dataset contains and what it represents
 - print the first 5 rows,
 - print basic information (use info()),
 - show descriptive statistics for numeric columns.

Task 3. Data integration

1. Create one consolidated DataFrame that describes each ride (one row = one ride) in a complete and readable way for analysis with the following data:
 - user name and user type,
 - bike type,
 - start station name and end station name,
 - ride duration,
 - payment amount.

2. Print the first and the last 10 rows.

Task 4. User activity analysis

1. Calculate:
 - total number of rides per user,
 - average ride duration per user.
2. Sort the results by number of rides (descending).
3. Display only users with at least 3 rides.

Task 5. Bike and station analysis

1. Calculate the average ride duration:
 - by bike type,
 - by district (based on start station).
2. Identify:
 - the most popular start station,
 - the most common bike type.

Interpret results briefly in Markdown.

Task 6. Financial analysis

1. Calculate:
 - total revenue,
 - average payment amount,
 - revenue by payment type.
2. Present the results in a clearly formatted table.

Task 7. User classification

1. Classify users into three groups based on total ride time:
 - low_activity
 - medium_activity
 - high_activity
2. Define your own thresholds and explain them in Markdown.

Task 8. Data visualization

1. Create the following charts:
 - Bar chart – number of rides by user type.
 - Histogram – distribution of ride duration.
 - Bar chart – total revenue by payment type.
2. Each chart must include:
 - title,
 - axis labels,
 - appropriate formatting.

Task 9. Station imbalance analysis

In a bike-sharing system, imbalance between incoming and outgoing rides causes operational problems (empty or overloaded stations).

1. For each station:
 - calculate the number of rides starting at the station,
 - calculate the number of rides ending at the station.
2. Create a new metric: $\text{imbalance} = \text{rides_started} - \text{rides_ended}$
3. Identify:
 - stations with the highest positive imbalance,
 - stations with the highest negative imbalance.
4. Classify stations into:
 - source (many departures),
 - sink (many arrivals),
 - balanced.

Thresholds must be defined and justified by the team.

Task 10. Revenue efficiency analysis

Not all rides generate revenue equally. Some generate higher income per minute of usage.

1. For each ride, calculate: $\text{revenue_per_minute} = \text{payment_amount} / \text{ride_duration}$
2. Analyze:
 - average revenue per minute by bike type,
 - average revenue per minute by user type.
3. Answer in Markdown:
 - Which bike type is the most revenue-efficient?
 - Which user group generates the highest value relative to time?
4. Results must be presented in:
 - one summary table,
 - one visualization (bar chart).

Task 11. Revenue efficiency analysis

Understanding usage patterns over time is crucial for planning system capacity.

1. Extract from `ride_date` at least one new time-based feature, e.g.:
 - weekday vs weekend,
 - early / mid / late month.
2. Compare:
 - number of rides,
 - average ride duration across the defined time categories.
3. Identify unusual patterns (outliers, anomalies) and describe them in Markdown.

PART II – MICROSOFT EXCEL (POWER QUERY & PIVOT TABLES)

1. Complete all tasks using Microsoft Excel.
2. Complete each task on a separate sheet.
3. At the top of each sheet, place the title of the task and its content.
4. Import all data in Power Query.
5. Use Power Query and Pivot Tables to analyze data.